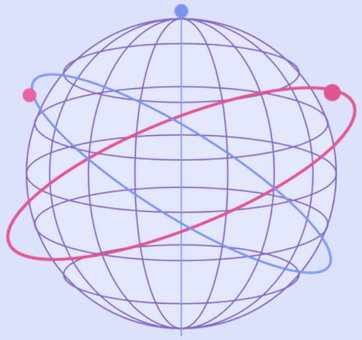
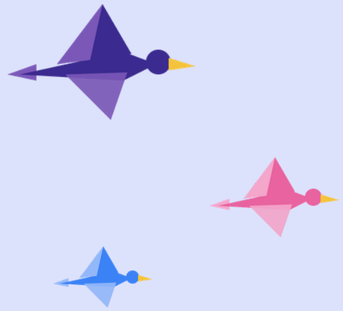
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
Integrated Dynamical Decoupling & Error Mitigation Pipeline for NISQ Workflows

TRACK
Track 3 - quantum hardware and error correction

TEAM NAME
QubitCrew

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Problem Statement

Integrated Dynamical Decoupling and Error Mitigation Pipeline for NISQ Workflows.

The Problem : on today's noisy devices, qubits sit idle while others run gates. Idle qubits dephase and pick up coherent ZZ crosstalk from their neighbours, so fidelity collapses long before the algorithm finishes.

01 THE GAP

- Applied separately and manually.
- Rigid, one-size-fits-all pulse sequences.
- Ignores specific idle window contexts.

02 OUR ROLE

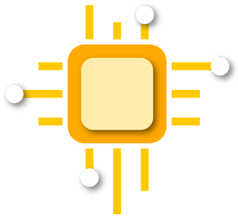
- Automated, hardware-aware pipeline.
- Detects and classifies idle windows.
- Optimizes DD placement and layers mitigation.
- Benchmarked against Qiskit & IBM Runtime.

03 DELIVERABLE

- ddpipe open-source Python package.
- Drop-in Qiskit transpiler pass & Streamlit app.
- Validated on real IBM Heron r2 hardware.

Objectives & Novelty

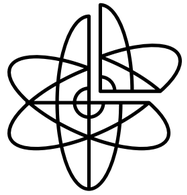
1 SMART IDLE DETECTION



100% AUTOMATED

- **Objective:** Detect idle windows directly from a scheduled circuit and label each window as idle-idle or crosstalk-idle.
- **Novelty:** Replaces manual idle management with automated context-aware tracking.

2 STAGGERED & AUTO DD



~54% INFIDELITY

- **Objective:** Insert DD trains that compose to identity and fit inside each window (XY4, uniform or staggered).
- **Novelty:** Staggered timing offsets neighbours to prevent ZZ-interaction accumulation; Auto mode minimizes ZZ-frame cost.

3 UNIFIED ERROR PIPELINE



3-IN-1 MITIGATION

- **Objective:** Combine dynamical decoupling with ZNE (CX folding) and readout-error mitigation in one unified pipeline.
- **Novelty:** Bridges separate mitigation techniques into a single, cohesive hardware workflow.

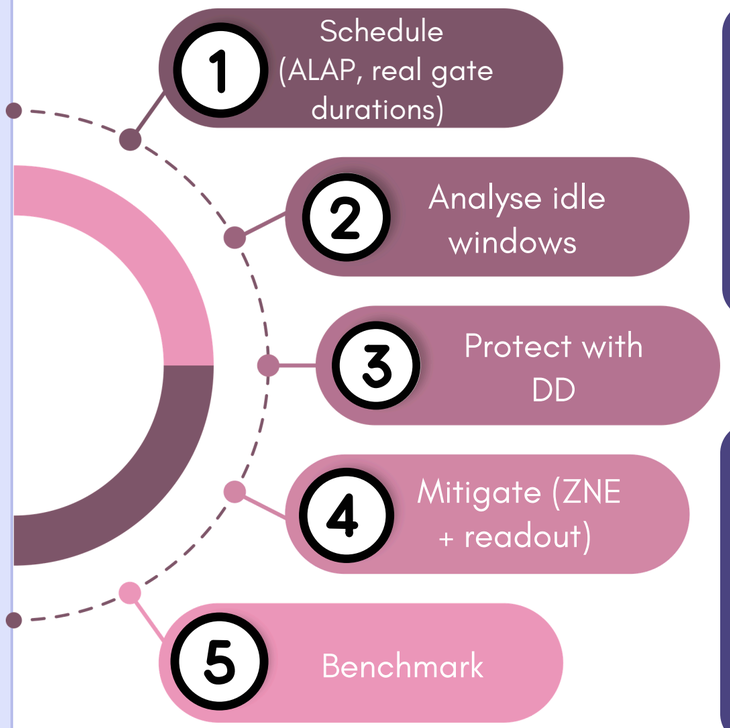
4 PRODUCTION-READY QISKIT



1-LINE PASS

- **Objective:** Benchmark fairly against Qiskit PadDD and IBM Runtime DD, in simulation and hardware.
- **Novelty:** “IdleAwareDD” packaged as a standard Qiskit transpiler pass requiring just one line in a PassManager.

Solution & Quantum Advantage



HARDWARE SPECTATOR TEST

- mean idle-qubit ⟨X⟩ rises from **-0.05** (no DD) to **0.78** with staggered DD on ibm_fez, and
- from **0.28** to **0.82** on ibm_kingston (ideal = 1).

SIMULATION

- auto-DD cuts mean infidelity from 0.281 to 0.129,
- a 54% reduction;
- Qiskit's DD reaches 0.142.

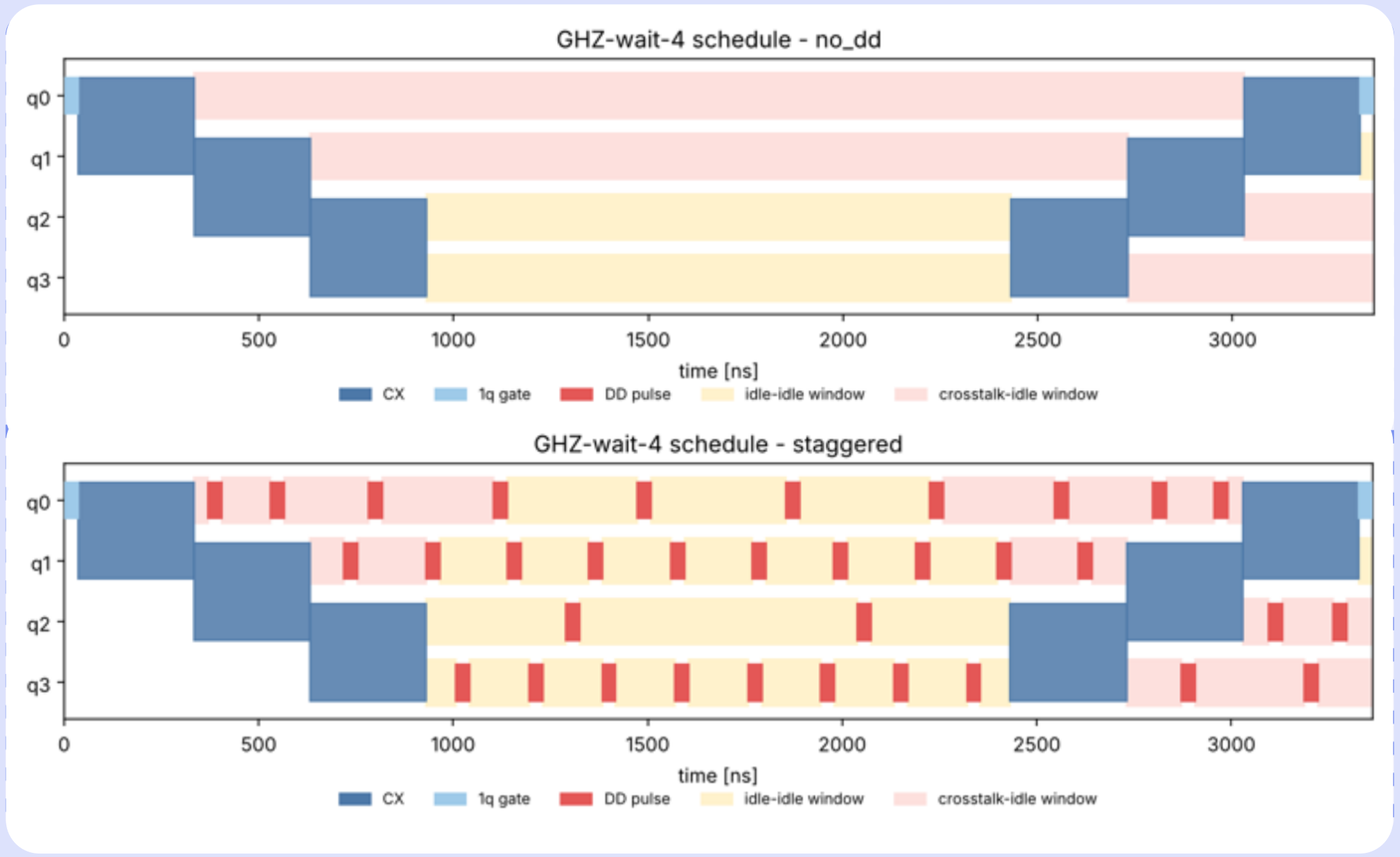
WHERE THE BENEFITS COME FROM?

long synchronous idle periods
e.g. spectator qubits waiting while a GHZ chain runs, and larger circuits such as a 6-qubit QAOA ring.

ADVANTAGE

we do not claim a computational speed-up.
The advantage is 'fidelity' : keeping idle-heavy circuits usable on today's noisy devices, at no extra qubits.

Algorithm & Circuit Design



KEY FEATURES

Circuits: GHZ-wait-4, QAOA-ring-6, VQE-TFIM-4.

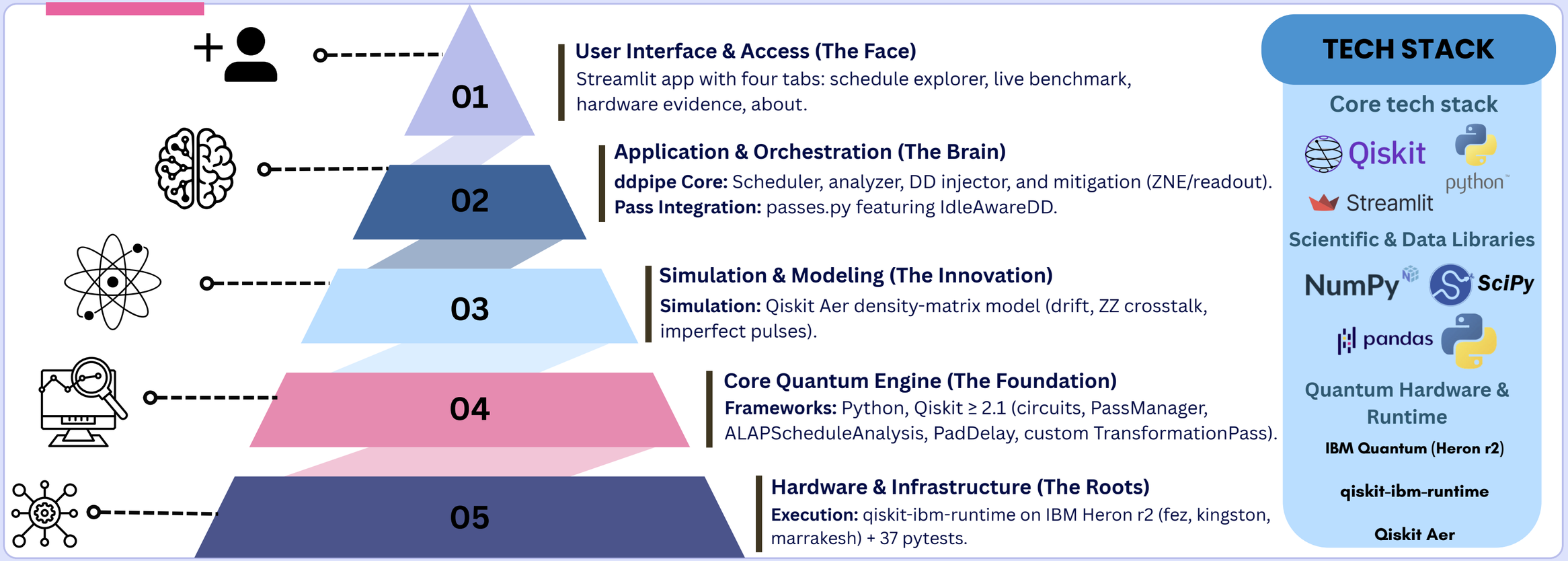
Schedule: transpile, ALAP schedule, PadDelay.

IdleAwareDD pass: XY4, τ = 600 ns, windows $\geqslant$ 250 ns, $\leqslant$ 16 pulses, uniform or staggered.

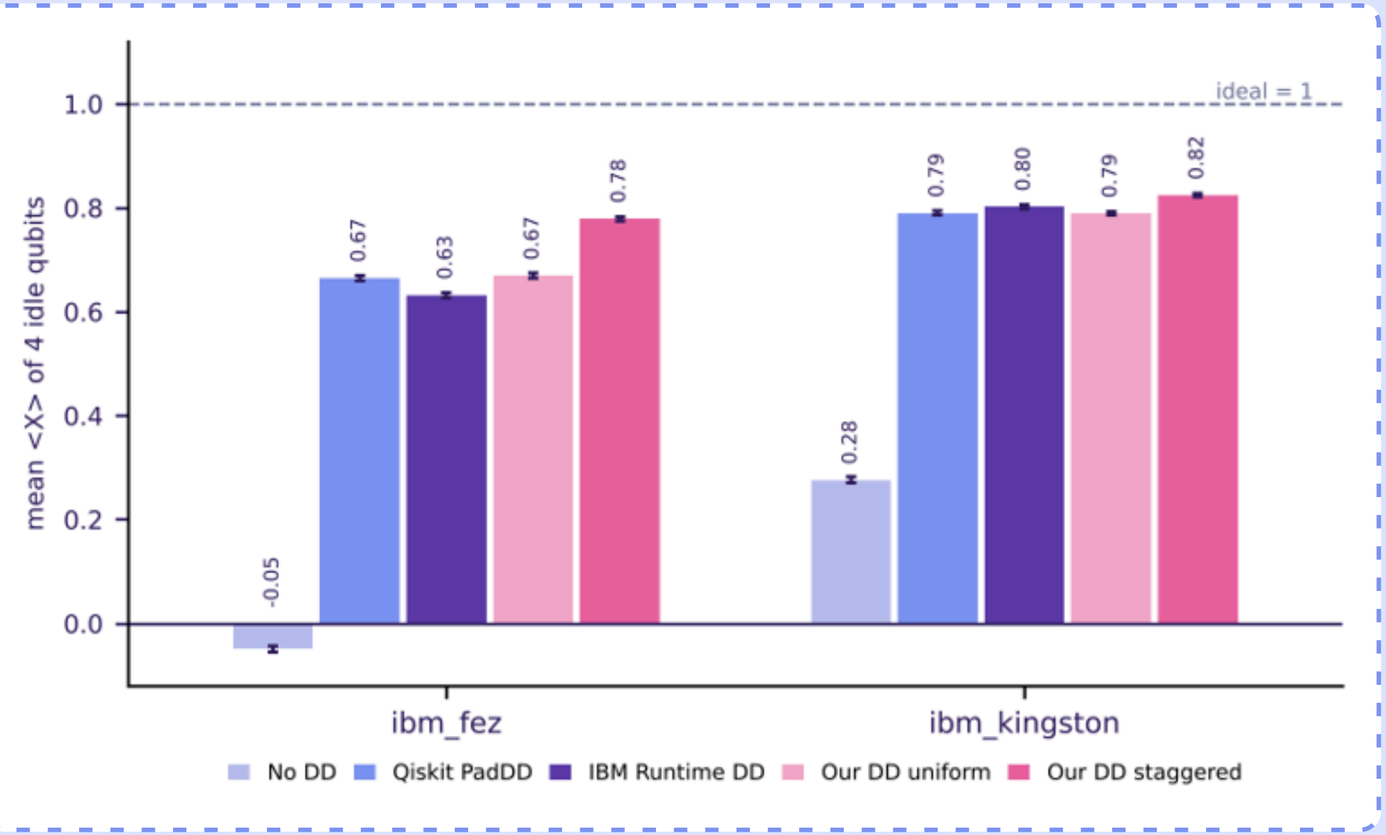
Mitigation: ZNE by CX folding (G G† G, scales 1, 2, 3) plus readout correction

Figure: red = DD pulse, yellow = idle-idle window, pink = crosstalk-idle window. Top: before DD. Bottom: staggered DD.

Implementation & Tech Stack



Results & Analysis



- **Staggered DD** is best on both devices, in all 3 interleaved repeats.
- **ibm_fez**: 0.78 vs Qiskit 0.67, IBM Runtime 0.63, our uniform 0.67, no DD -0.05.
- **ibm_kingston**: 0.82 vs 0.79, 0.80, 0.79; no DD 0.28. Margin over IBM's DD is small.
- **Crosstalk test (fez)**: staggered 0.82 vs uniform 0.72.
- **Simulation:** mean infidelity 0.281 → 0.129 (-54%); Qiskit DD 0.142.
- **Metric:** mean ⟨X⟩ of 4 idle qubits, ideal = 1; error bars are standard errors.

Impact & Real-World Applications

01
Who benefits

anyone running circuits with idle time on NISQ hardware: variational algorithms (VQE, QAOA), state preparation and circuits with mid-circuit waits.

02
Lower barrier

IdleAwareDD drops into an existing Qiskit PassManager, so DD no longer needs hand-built sequences.

03
Fair benchmarking

one tool that compares no DD, Qiskit DD, IBM Runtime DD and ours on the same circuits, with the raw hardware job results committed.

04
Useful today

better fidelity from existing devices, with no extra qubits or hardware changes, which helps near-term chemistry and optimisation experiments reach deeper circuits.

05
Education & research

the schedule explorer shows exactly which windows get which pulses, making DD behaviour easy to teach and inspect.

Challenges & Future Scope

CHALLENGES	**FREE-PLAN QPU TIME LIMITS**	**DD PLUS ZNE PERFORMANCE**	**SPECTATOR-CIRCUIT WINDOW CLASSIFICATION**
DETAILS	Repeats hardware baselines drift between sessions, so we compare only within a run	Did not beat Runtime's ZNE alone on ibm_marrakesh (0.74 vs 0.80), and the margin over IBM's DD on ibm_kingston is small	Every window was classified idle-idle; results support the injector and staggering more than the crosstalk-idle label. Busy-qubit fidelity is dominated by CZ errors

FUTURE SCOPE: **More repeats and devices; per-qubit ZZ rates read from backend calibration; richer sequences (UDD, CDD); dynamic circuits; tighter integration with Runtime resilience options.**

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.

GITHUB REPOSITORY
https://github.com/vtu-qiskit-fall-fest-2026/vtu-hc-qubitcrew

DEMO VIDEO LINK
https://youtu.be/cA8_7WGutxc

